

aggressive in times of favorable price momentum and less aggressive in times of adverse price momentum and might give the impression that trading at faster rates could lead to lower costs than trading at a slower rate. We only had access to the actual trading conditions and market volumes over the trading horizon.

- Third, we did not discard data points or stocks. Readers and analysts should remember that our research step is a learning step. All data points should be included and observed to fully understand the underlying data set and system at hand—including outliers.
- Fourth, we require at least 25 data points for each bucket; less than 25 data points does not provide a reasonable point estimate of the trade cost for that particular interval. Unfortunately, the fourth requirement resulted in largely smaller trade sizes.
- Fifth, we did not know if these shares were part of a larger order where parts were executed the previous day(s) and/or portions were to be executed on subsequent day(s).
- Finally, the  $R^2$  statistic reported is the  $R^2$  for the grouped and averaged data set. In these cases, the measure is often inflated—however, it does provide insight into whether the dependent variables are related to the independent variable (e.g., trading cost). Here we are still in our learning mode—so the  $R^2$  on grouped data indeed provides valuable insight.

Our graphical illustrations contained all data points. We computed buckets for the x-value and then computed the average cost for all data points that fell into that particular bucket. For example, we computed the average cost for all order sizes that were 5% ADV (rounded to nearest 1% ADV). The data points included in these intervals may have varying volatilities and POV rates as well as prices. All of these variables will have some effect on the actual cost of the trade. However, in this research and learn step, even this type of averaging approach will yield some insight into the underlying relationships between cost and variable.

*Large Cap Stocks.* Our large cap observations are shown in [Figure 5.1](#). Findings mirror results in leading academic research. Costs were positively related to size, volatility, and POV rate, and negatively related to price. We ran a simple linear regression on the grouped data for each explanatory variable separately. The strongest relationship (based on  $R^2$ ) for cost was size ( $R^2 = 0.44$ ). The second strongest relationship was with volatility ( $R^2 = 0.40$ ). This was followed by POV rate ( $R^2 = 0.36$ ) and then price ( $R^2 = 0.24$ ). In each of these cases, visual inspection shows that the relationship between cost and explanatory variable may be non-linear.

*Small Cap Stocks.* Our small cap observations are illustrated in [Figure 5.2](#). Our findings also reflected the academic research with large